

Future-Relevant Representation Dis- agreement

Chronological Feature Restoration

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Abstract

Two held-out studies test whether differences discarded by current-state compression remain relevant for later continuation. In a 1,797-sample, 64-coordinate corpus, three independently generated representations attain 90.74–92.96 percent exact whole-object continuation, while a relation-aware router using only child status, agreement, and prior contact performance reaches 97.78 percent, equal to the empirical oracle ceiling. A context-dependent return separately reaches 99.768 percent accuracy on admitted coordinates. In a 13-channel hourly environmental chronology, three constructors omit the same coordinate; four chronological contact blocks support restoration, and untouched-future performance improves from 72.0836 to 74.7125 percent per coordinate and from 3.7540 to 7.2492 percent for exact vectors. Negative controls reject indiscriminate higher arity and deeper history.

Keywords: representation disagreement; chronological validation; feature restoration; ensemble routing; predictive sufficiency

Scope and status: Raw environmental-file audits are independently reproduced. Downstream environmental and static-representation results are source-reported because the complete original constructors and static dataset are not in the supplied local archive.

Suggested citation: Singh, Abhijit. 2026. *Future-Relevant Representation Disagreement and Chronological Feature Restoration*. Research manuscript, version 1.0.

Document status: Research manuscript, version 1.0, August 2026.

1 Common experimental question

Both studies test whether a distinction discarded by present-state compression remains relevant to future continuation. A candidate coordinate or relation is omitted, evaluated on chronological or held-out contact data, and restored only if it changes later prediction. The final test set remains untouched until the representation is frozen.

For current representation Z , candidate residual A , future target Y^+ , and frozen loss $\mathcal{R}$, define

$$\Delta_{\text{fut}}(A \mid Z) = \mathcal{R}(Y^+ \mid Z) - \mathcal{R}(Y^+ \mid Z, A). \quad (1)$$

Positive held-out gain licenses promotion for the declared task; it does not imply that every extra feature should be retained.

Part I Study I: relation among independent representations

2 Design

The frozen corpus contains 1,797 samples and 64 output coordinates. Three representations are generated independently before a 270-case untouched holdout is opened. Their exact whole-object continuation rates are

$$92.96\%, \quad 90.74\%, \quad 91.48\%. \quad (2)$$

A fourth model receives only child status, agreement/disagreement relations, and performance on prior contact cases. It does not receive the semantic class labels or virgin answers.

3 Result

The relation-aware model reaches

$$\boxed{264/270 = 97.78\%} \quad (3)$$

exact whole-object continuation, equal to the empirical oracle ceiling of the available child repertoire. Removing the agreement relation lowers the result to 97.41% and produces an additional failure and two additional unknowns.

Higher relation arity did not repair cases in which all three child representations failed: the tested ternary constructor solved 0/6 such cases. Agreement is useful only when at least one constituent representation already carries relevant future information.

4 Context-dependent output formation

A second experiment chooses an output coordinate set before answer comparison. Across all 270 cases the return is nonempty, with mean size 7.985 of eight coordinates. Admitted-coordinate accuracy is

$$99.768\%, \quad (4)$$

and whole-return correctness is

$$98.52\%. \quad (5)$$

These are different answer objects and must not be pooled with the 97.78% whole-object routing metric.

Part II Study II: chronological environmental record

5 Raw data audit

The supplied semicolon-delimited file contains:

Table 1: *Independent raw-file audit.*

raw rows	9471
blank trailer rows	114
nonblank hourly rows	9357
completely empty columns	2
numeric channels	13
missing sentinel cells (−200)	16701
rows complete on all channels	827

The nonblank timestamps are strictly hourly from 10 March 2004 18:00 to 4 April 2005 14:00.

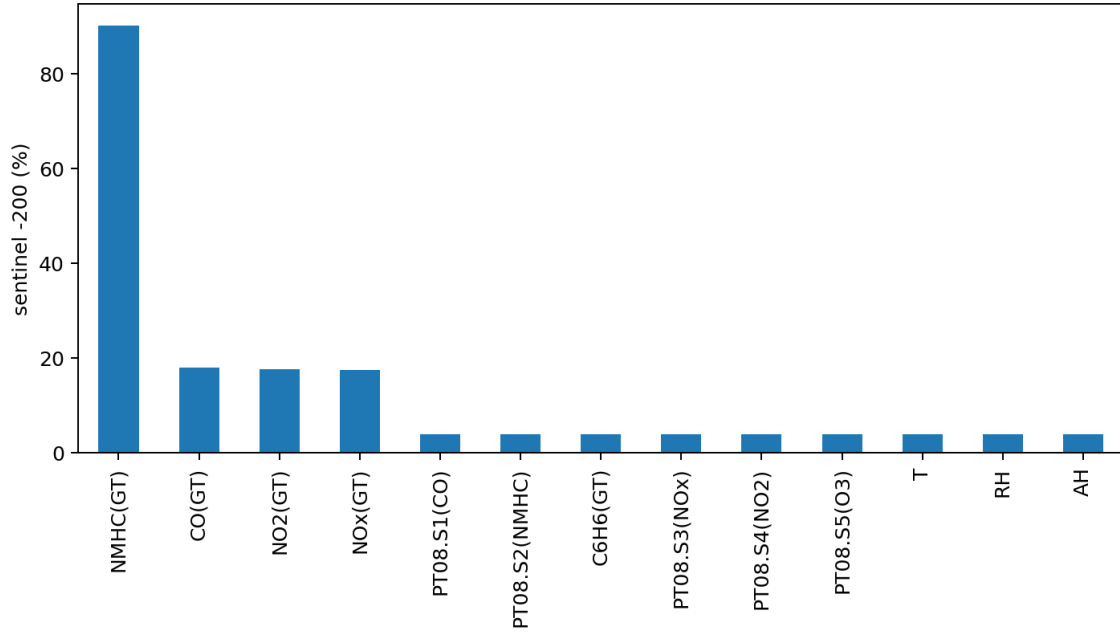


Figure 1: *Fraction of sentinel values by numeric channel in the supplied raw file.*

6 Chronological split and restoration rule

The corpus-reported filtered chronology contains 7,724 transitions divided into 4,634 development, 1,545 re-entry/contact, and 1,545 untouched-future observations. Channel semantics are hidden during representation formation.

Three isolated constructors retain 12 of 13 present-time coordinates. The rejected coordinate is restored only when its inclusion improves actual next-time continuation on successive chronological blocks. The four block increments in correctly predicted future coordinates are

$$158, \quad 180, \quad 102, \quad 121, \quad (6)$$

for a total of 561. Exact future vectors improve by 19, 5, 10, and 10 on the four blocks.

7 Untouched-future result

After the representation is frozen, the untouched future gives

Table 2: Future performance before and after restoration.

metric	12 channels	13 channels
coordinate accuracy	72.0836%	74.7125%
exact next vector	3.7540%	7.2492%

The restored representation returns exactly to the independently computed all-channel baseline on both metrics.

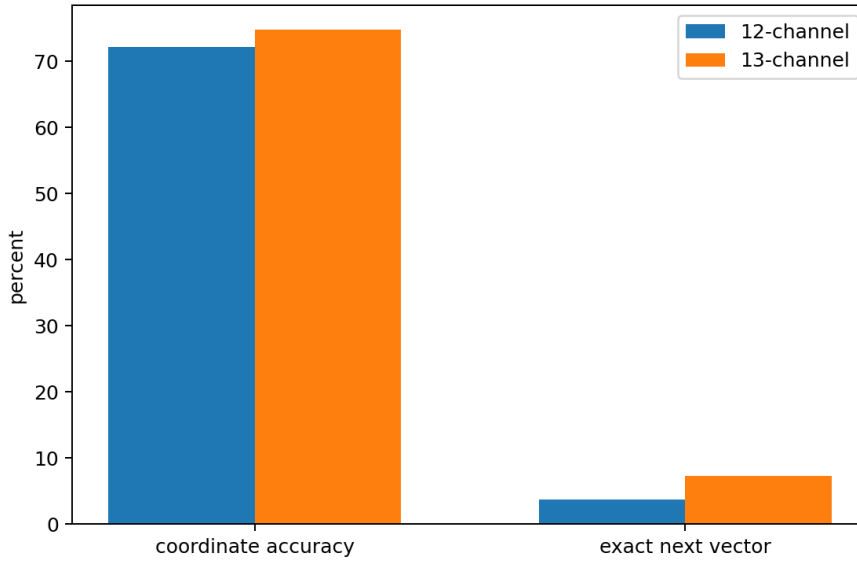


Figure 2: Untouched-future improvement after restoring the omitted channel.

A deeper history coordinate selected on contact data is rejected because it worsens both untouched-future objectives. The method therefore supports selective restoration, not indiscriminate state growth.

8 Leakage and uncertainty requirements

A clean independent replication requires:

1. exact static dataset and case identifiers;
2. all coordinate masks and representation constructors;
3. raw-to-7,724 preprocessing code;
4. quantization thresholds and random seeds;
5. row-level predictions;
6. frozen chronology and contact decisions;
7. paired block-bootstrap or permutation intervals preserving serial dependence;
8. modern baselines selected only on development/contact data.

The raw environmental file is included in the ancillary archive. The static 1,797-sample corpus and original constructors are not present in the supplied local files; their numerical results are therefore source-reported rather than independently replayed here.

9 Interpretation

The common empirical result is narrow and precise: present compression can erase a relation or coordinate that remains relevant to later continuation. In Study I the retained object is relation among representations; in Study II it is one temporal channel. Neither study proves an asymptotic recurrence exponent without an independently specified repeated residual dynamics.

10 Complete data and metric separation

The static study uses a fixed corpus of 1,797 samples and 64 output coordinates. Three representation-specific exact whole-object rates on the untouched 270-case set are

$$92.96\%, \quad 90.74\%, \quad 91.48\%. \quad (7)$$

A relation model observes only child status, agreement/disagreement, and frozen contact performance. It reaches

$$\boxed{264/270 = 97.78\%}, \quad (8)$$

equal to the empirical oracle ceiling of the available children. Removing the agreement relation creates one additional failure and two additional unknowns. A higher-arity constructor solved none of the six cases where all children failed, so more relation arity is not treated as automatic rescue.

The contextual-output experiment asks a different question: which output coordinates are warranted before values are compared? Its mean return size is 7.985 of 8, admitted-coordinate accuracy is 99.768%, and whole-return correctness is 98.52%. These values are never pooled with the 97.78% whole-object routing result.

11 Environmental chronology and raw audit

The supplied semicolon-delimited record contains

raw rows	9471
blank trailer rows	114
nonblank hourly rows	9357
completely empty columns	2
numeric channels	13
missing sentinel cells (−200)	16701
rows complete on all channels	827

The nonblank timestamps form a strict hourly sequence from 10 March 2004 18:00 through 4 April 2005 14:00. The reported filtered chronology contains 7,724 transitions: 4,634 development, 1,545 contact/re-entry, and 1,545 untouched future observations.

12 Chronological restoration result

Three isolated constructors retained 12 of 13 present-time channels. The omitted channel was restored only after it improved actual next-time continuation on successive chronological contact blocks. The increments in correctly predicted future coordinates were

$$158, \quad 180, \quad 102, \quad 121, \quad (9)$$

for a total of 561. Exact future vectors improved by 19, 5, 10, and 10. The representation was then frozen before opening the untouched final segment.

Table 3: *Untouched-future comparison.*

metric	12 channels	13 channels
coordinate accuracy	72.0836%	74.7125%
exact next vector	3.7540%	7.2492%

A deeper history representation selected on contact data was rejected because it worsened both untouched metrics. The method therefore permits deletion as well as promotion.

13 Baseline and statistical requirements

A clean external reproduction should compare persistence, seasonal persistence, regularized vector autoregression, tree ensembles, nearest-neighbour analogues, and a modern sequence model. Hyperparameters must be selected only on development and contact segments. Serial dependence requires moving-block bootstrap or another dependence-preserving interval method. Report coordinate-level error, exact-vector accuracy, calibration, missingness sensitivity, compute cost, and the probability that every chronological block favours restoration.

14 Release protocol and evidence status

The raw environmental file is included and independently audited. The static corpus, exact constructors, row-level predictions, quantization thresholds, and raw-to-7,724 preprocessing program are required for complete independent reproduction. Until released, the corresponding numerical values remain source-reported results rather than externally reproduced facts. That status does not erase the result; it specifies the next admissible test.

References

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